COMPUTER NETWORKS

Unit 2 - Complete Notes

Routing Algorithms • IP Addressing (IPv4 and IPv6) • Subnetting and Supernetting • DHCP • VLAN • Congestion Control • Tunnelling • Fragmentation

Chapter	Syllabus Topic	Included
2.1	Routing Algorithm - Network layer design issues, classful and classless IP addressing, subnetting, supernetting, forwarding of IP packets, IPv4 header, IPv4 and IPv6 addressing, network layer protocols	Yes
2.2	Distance vector, link state, hierarchical and broadcast routing, shortest path algorithm, unicast routing, DHCP, VLAN	Yes
2.3	Congestion prevention policies, congestion control in datagram subnets, load shedding, jitter control, leaky bucket and token bucket algorithms, tunnelling, internetwork routing, fragmentation	Yes

How to use these notes: theory first, then the worked examples - subnetting, Dijkstra, fragmentation and token bucket are the most frequently asked numericals from this unit. Tables and comparison charts are written for last-minute revision.

Chapter 2.1 - Routing Algorithm, IP Addressing and Network Layer Protocols

1. Network Layer Design Issues

The network layer is responsible for delivering packets from the source host to the destination host, possibly across multiple heterogeneous networks. Its main design issues are:

- **Routing:** choosing the best path (route) from source to destination. Done by routing algorithms, which fill routing tables.
- **Forwarding:** moving a packet from the router's input port to the correct output port using the routing table. (Routing = filling the table, forwarding = using it.)
- **Logical addressing:** every host gets a unique IP address, independent of the underlying hardware (MAC) address.
- **Fragmentation and reassembly:** if a packet is larger than the network's MTU, it is split into fragments and reassembled at the destination.
- **Congestion control:** preventing too many packets from flooding the subnet (covered in Chapter 2.3).

Services provided to the transport layer should be:

- Independent of the subnet topology (router count, layout).
- Independent of the technology of the routers (no transport layer changes if routers are replaced).
- Uniform addressing scheme - a transport entity should be able to talk to any other, anywhere.

Connectionless (datagram) vs Connection-oriented (virtual circuit) subnets

Issue	Datagram Subnet	Virtual Circuit Subnet
Setup phase	None - packets sent anytime	Connection (VC) established first; all packets follow it
Address carried	Full source + destination address in every packet	Short VC identifier in each packet
State in routers	None - stateless	Each VC occupies a table entry
Routing	Each packet routed independently (may take different paths)	All packets follow the same fixed route
Failure handling	Reroute around failed router easily	VC breaks; connection must be re-established
Congestion control	Difficult (no reserved resources)	Easier - resources can be reserved in advance
Examples	IP (the Internet)	ATM, MPLS, X.25, frame relay

2. IP Addressing - IPv4

An IPv4 address is **32 bits**, written as four decimal octets separated by dots (dotted-decimal notation), e.g. 192.168.10.25. Every address has a **network part** and a **host part**.

2.1 Classful Addressing (original scheme)

Class	Leading bits	1st octet range	Default mask	Max hosts per network	Use
A	0	1 - 126	255.0.0.0 (/8)	16,777,214	Very large networks
B	10	128 - 191	255.255.0.0 (/16)	65,534	Medium networks
C	110	192 - 223	255.255.255.0 (/24)	254	Small networks
D	1110	224 - 239	-	-	Multicast groups

Class	Leading bits	1st octet range	Default mask	Max hosts per network	Use
E	1111	240 - 255	-	-	Experimental / reserved

- **Class A:** first octet = network, last 3 octets = host. Only 126 networks but huge host count - wasteful for small users.
- **Class B:** first two octets = network, last two = host.
- **Class C:** first three octets = network, last octet = host - max 254 hosts per network.
- **Special addresses:** 127.x.x.x = loopback (test local stack); host part all 0s = network address; host part all 1s = directed broadcast; 0.0.0.0 = 'this host' (source at boot); 255.255.255.255 = limited broadcast.
- **Private addresses (RFC 1918, not routable on Internet):** 10.0.0.0 - 10.255.255.255, 172.16.0.0 - 172.31.255.255, 192.168.0.0 - 192.168.255.255; 169.254.x.x = APIPA (self-assigned when DHCP fails).

2.2 Classless Addressing (CIDR - Classless Inter-Domain Routing)

- Classful addressing wasted addresses and made routing tables explode. CIDR removed the classes.
- Written as **a.b.c.d/n** where n = number of network bits (the prefix length), e.g. 192.168.4.0/24 means the first 24 bits identify the network.
- The mask has n ones followed by (32 - n) zeros, e.g. /26 = 255.255.255.192.
- Advantage: address blocks can be allocated in exact sizes needed, and many small networks can be aggregated (summarized) into one route - this reduces router table size.

2.3 Subnetting

Idea: 'borrow' bits from the host part of an address to create smaller sub-networks inside one network. Subnetting reduces broadcast traffic, improves security and stops address waste.

- **Steps:** (1) find how many subnet/host bits are needed; (2) write the new mask; (3) the block size = 256 - mask's interesting octet value; (4) enumerate network addresses, host ranges and broadcast addresses.
- **Number of subnets** = 2^(borrowed bits); **usable hosts per subnet** = 2^(remaining host bits) - 2 (network and broadcast are reserved).

Worked Example 1: Subnetting 192.168.1.0/24 into 4 subnets

Need 4 subnets: 2^2 = 4, so borrow 2 bits -> prefix /26, mask 255.255.255.192, block size 64, 62 usable hosts each.

Subnet	Network address	Host range	Broadcast address
1	192.168.1.0	192.168.1.1 - 192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65 - 192.168.1.126	192.168.1.127
3	192.168.1.128	192.168.1.129 - 192.168.1.190	192.168.1.191
4	192.168.1.192	192.168.1.193 - 192.168.1.254	192.168.1.255

Worked Example 2: Subnetting for hosts

Given 172.16.0.0/16, a department needs 500 hosts per subnet. 2^9 - 2 = 510 (works), so 9 host bits -> prefix /23, mask 255.255.254.0, block size 2 in the third octet. Subnets: 172.16.0.0/23, 172.16.2.0/23, 172.16.4.0/23, ...

Quick subnetting reference

Prefix	Mask	Usable hosts	Prefix	Mask	Usable hosts
/24	255.255.255.0	254	/27	255.255.255.224	30
/25	255.255.255.128	126	/28	255.255.255.240	14
/26	255.255.255.192	62	/30	255.255.255.252	2 (links)

2.4 Supernetting (Route Aggregation)

- **Idea:** the reverse of subnetting - combine several consecutive classful networks into one larger block, using a shorter prefix.
- **Example:** 192.168.0.0/24, 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24 (four Class C networks) can be advertised together as **192.168.0.0/22** - one entry instead of four, because the first 22 bits are common.
- Reduces routing table size on the Internet (this summarization is a key benefit of CIDR).

3. Forwarding of IP Packets

- Each router keeps a **routing table** with entries: destination network/prefix, subnet mask, next-hop router (or 'direct'), outgoing interface, and optionally a metric.
- When a packet arrives, the router **ANDs** the destination IP with each entry's mask; if it matches the entry's network, the packet is forwarded to that entry's next hop.
- **Longest prefix match:** if several entries match, the one with the longest (most specific) prefix wins - this is how CIDR and supernetting work together.
- Default route: 0.0.0.0/0 - used when no other entry matches (gateway of last resort).

Exam tip: a typical numerical gives you a routing table and a few destination IPs - AND each IP with each mask, apply longest prefix match, and state the next hop.

4. IPv4 Header

The IPv4 header is **20 bytes minimum, up to 60 bytes** with options. Fields (in order):

Field	Size	Purpose
Version	4 bits	IP version in use (4)
IHL (Header Length)	4 bits	Header length in 32-bit words: 5 to 15, i.e. 20 to 60 bytes
DSCP / Type of Service	8 bits	Priority / QoS treatment for the packet
Total Length	16 bits	Header + data length in bytes (max 65,535)
Identification	16 bits	Same value in all fragments of one datagram (for reassembly)
Flags	3 bits	DF = Do not Fragment; MF = More Fragments follow
Fragment Offset	13 bits	Position of this fragment's data, in units of 8 bytes
TTL (Time To Live)	8 bits	Hop counter; decremented each router; packet discarded at 0 (prevents loops)
Protocol	8 bits	Upper-layer protocol: 6 = TCP, 17 = UDP, 1 = ICMP
Header Checksum	16 bits	Integrity check on the header only; recomputed at every hop (TTL changes)
Source Address	32 bits	Sender's IP address
Destination Address	32 bits	Receiver's IP address
Options	0 - 40 bytes	Rarely used: record route, timestamp, strict/loose source route

5. IPv6 Addressing

- IPv6 addresses are **128 bits**, written as 8 groups of 4 hexadecimal digits separated by colons, e.g. 2001:0db8:0000:0000:0000:ff00:0042:8329.
- Leading zeros in a group may be omitted; **one** run of consecutive zero groups may be replaced by '::'. Example above -> 2001:db8::ff00:42:8329.

IPv6 header (fixed 40 bytes, no options field)

Field	Size	Purpose
Version	4 bits	6
Traffic Class	8 bits	QoS / priority (like DSCP)
Flow Label	20 bits	Marks packets of one flow for special handling
Payload Length	16 bits	Length of data after the 40-byte header
Next Header	8 bits	Type of the next header: an extension header or upper protocol (6 = TCP, 17 = UDP, 58 = ICMPv6)
Hop Limit	8 bits	Same job as IPv4 TTL
Source Address	128 bits	Sender
Destination Address	128 bits	Receiver

IPv4 vs IPv6 - key differences

Feature	IPv4	IPv6
Address size	32 bits	128 bits (huge space; restores end-to-end addressing)
Header	20 - 60 bytes, 14 fields	Fixed 40 bytes, 8 fields - faster router processing
Header checksum	Present; recomputed every hop	Removed (lower layers already check integrity)
Fragmentation	Routers and source can fragment; reassembly at destination	Only the SOURCE fragments; routers never fragment (simpler routers)
Options	Variable part of header	Extension headers placed between header and payload
Broadcast	Supported	Not supported - replaced by multicast and anycast
Configuration	Manual or DHCP	Autoconfiguration (SLAAC) + DHCPv6
Address types	Unicast, broadcast, multicast	Unicast (global, link-local FE80::/10), multicast (FF00::/8), anycast

- **Anycast (new in IPv6):** one address assigned to several hosts; a packet to it is delivered to the NEAREST one - used for load balancing and finding the closest server.

6. Network Layer Protocols (working and purpose)

Protocol	Full name	Working / purpose
IP	Internet Protocol	Delivers datagrams between hosts; connectionless, best-effort, no guarantees
ICMP	Internet Control Message Protocol	Error reporting and diagnostics between hosts/routers (destination unreachable, time exceeded); ping uses Echo Request / Echo Reply
ARP	Address Resolution Protocol	Finds the MAC address of a host on the same LAN, given its IP (request broadcast, reply unicast)
RARP	Reverse ARP	Finds the IP given the MAC - superseded by BOOTP and DHCP
IGMP	Internet Group Management Protocol	Lets hosts join / leave multicast groups; routers use it to track group members on each interface

Protocol	Full name	Working / purpose
DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP address, mask, gateway and DNS to hosts at boot time (detailed in Chapter 2.2)
NAT	Network Address Translation	Maps many private addresses to one (or few) public IP(s) at the router - conserves IPv4 addresses

Chapter 2.2 - Routing Algorithms, DHCP and VLAN

1. Shortest Path Algorithm (Dijkstra)

- **Goal:** find the least-cost path between a source and every other node in the network, where cost can be hops, distance, delay or a money measure.
- **Method:** each node is labelled with its best-known distance from the source; the node with the smallest tentative label is made permanent, and its neighbours are relaxed (updated). Repeat until all nodes are permanent.

Worked example

Topology and link costs: A-B = 2, A-C = 1, B-C = 2, B-D = 1, C-D = 4, C-E = 3, D-E = 1. Find shortest paths from A.

Step	Permanent set N'	B	C	D	E
0	A	2 (via A)	1 (via A)	infinity	infinity
1	A, C	2 (via A)	-	5 (via C)	4 (via C)
2	A, B, C	-	-	3 (via B)	4 (via C)
3	A, B, C, D	-	-	-	4 (tie: via C or via D)
4	all	-	-	-	-

Result: A->C = 1, A->B = 2, A->B->D = 3, A->C->E = 4 (or A->B->D->E = 4).

Complexity: O(n^2) naively; can be reduced with heap data structures. Used inside link state routing (OSPF).

2. Distance Vector Routing (DVR)

- Used by **RIP**. Each router keeps a table with the **best known distance to every destination and the next hop** - a vector of distances.
- Routers periodically exchange their entire tables with **directly connected neighbours only**.
- On receiving a neighbour's table, a router applies the **Bellman-Ford update**: D_x(y) = min over all neighbours v of [c(x,v) + D_v(y)] - if a cheaper route via that neighbour is found, update the entry.

Worked example: line network A - B - C - D (each link cost 1)

Router A learns (final)	Distance	Next hop	How it was learned
A to B	1	B	direct neighbour
A to C	2	B	C is 1 from B (1 + 1)
A to D	3	B	D is 2 from B (2 + 1)

Count-to-infinity problem

- If a link goes down, two neighbours can keep telling each other 'I can reach X through you' - bad news travels slowly; the estimated distance rises to infinity (16 = infinity in RIP).
- **Fixes:** split horizon (never advertise a route back to the neighbour you learned it from), split horizon with poison reverse (advertise it back with cost infinity), and hold-down timers (ignore updates about a failed route for a while).

3. Link State Routing (LSR)

- Used by **OSPF** and IS-IS. Every router learns the **complete map** of the network and runs Dijkstra on it.

Five steps

- **1. Discover neighbours:** send HELLO packets on each link; record who replies.
- **2. Measure cost:** measure delay / bandwidth to each neighbour (echo packets with timestamps).

- **3. Build Link State Packet (LSP):** containing (router ID, neighbours, costs, sequence number, age).
- **4. Distribute LSPs by flooding:** each router forwards the LSP on all links except the one it arrived on, using sequence numbers to reject duplicates/loops (acknowledged for reliability).
- **5. Compute shortest paths:** once every router has all LSPs, it reconstructs the full topology and runs Dijkstra to build its routing table.

Distance Vector vs Link State

Point	Distance Vector	Link State
Knowledge	Only distance to destinations, via neighbours' summaries	Full topology map of the whole network
Information shared	Entire routing table with neighbours only	LSPs flooded to all routers
Algorithm	Bellman-Ford (distributed)	Dijkstra (run locally on complete map)
Update trigger	Periodic (every 30 s in RIP) plus triggered updates	On change only (faster convergence)
Bandwidth use	Small messages but frequent	Heavy at start / on changes only
Convergence	Slow; count-to-infinity problem	Fast; loops less likely
Robustness	Bad information propagates	One router's error affects only itself
Examples	RIP	OSPF, IS-IS

4. Hierarchical Routing

- As networks grow, routing tables become huge. **Idea:** divide routers into **regions**; a router knows full detail about its own region but treats other regions as a single destination.
- Trade-off: smaller tables and less update traffic, at the cost of slightly longer routes than optimal.
- For n routers, the optimal number of hierarchy levels is approximately **ln n**, giving average table size of about e . ln n entries.
- This is exactly how the Internet works: networks, then areas, then autonomous systems (domains).

5. Broadcast Routing

Delivering a packet from one source to **all** hosts. Four techniques:

- **Flooding:** send on every outgoing link except the one it arrived on. Simple and robust but generates duplicate packets - controlled by a **sequence number per source**: a router forwards a packet only if it has not seen that sequence number before.
- **Multi-destination routing:** the packet carries a list of destinations; each router picks out the destinations reachable through each outgoing line, splits the list and forwards - one copy per needed link.
- **Reverse Path Forwarding (RPF):** a broadcast packet is forwarded only if it arrives on the link that lies on the shortest path back to the source - eliminates most duplicates cheaply, one copy per LAN.
- **Spanning-tree routing:** build a spanning tree covering all LANs; each packet is forwarded only along tree branches - the most efficient scheme, one copy per link (basis of multicast routing).

6. Unicast Routing

- Delivery of a packet from **one source to one destination** - the normal case handled by all the algorithms above (DVR, LSR, Dijkstra).
- Destination-based forwarding: the router looks only at the destination address in the routing table - one table lookup per packet, independent of the source.

7. DHCP - Dynamic Host Configuration Protocol

- **Purpose:** automates IP configuration. A host boots with no IP and gets one automatically: address, subnet mask, default gateway, DNS server(s) and the lease time - all in one transaction.

- **Transport:** UDP; server port 67, client port 68. Works across routers via a DHCP relay agent.

DORA - the four-step exchange

Step	Message	Who sends	What happens
1	DHCP Discover	Client -> broadcast	'I need an IP' (to 255.255.255.255, since client has no address)
2	DHCP Offer	Server -> client	One (or more) servers offer an address + lease parameters
3	DHCP Request	Client -> broadcast	Client accepts one offer (broadcast so other servers withdraw theirs)
4	DHCP Acknowledge	Server -> client	Lease confirmed; client configures the address and starts the timer

- **Allocation methods:** automatic (permanent), dynamic (leased, returned when host leaves - reuses scarce addresses), manual (admin fixed).

8. VLAN - Virtual LAN

- **Definition (IEEE 802.1Q):** a group of devices that communicate as if on the same LAN, regardless of their physical location - membership is configured in the switch, not by cabling.
- Switches tag frames with a **4-byte VLAN ID** (inserted after the source address); switches strip or add tags between trunk links.

Benefits

- Each VLAN is a separate **broadcast domain** - broadcast traffic stays inside the VLAN (better performance).
- **Security and management:** departments are isolated; users can only reach services of their own group.
- **Flexibility:** moving a user to another department = a configuration change, no rewiring.
- **Types:** port-based (switch port membership - most common), MAC-based (by device address), and protocol-based.

Chapter 2.3 - Congestion Control, Tunnelling and Fragmentation

1. Congestion - what and why

- **Congestion:** too many packets in (part of) the subnet - demand exceeds capacity, queues grow, delay rises sharply, and routers start dropping packets.
- **Causes:** sudden bursts of traffic, too many hosts sending at full speed, slow processors/links, insufficient memory (buffer overflow), retransmissions of lost packets making it worse (congestion collapse).
- Congestion is a **global** phenomenon involving many hosts and routers; flow control (one sender, one receiver) cannot solve it.

2. Congestion Prevention Policies

Open-loop (prevention): act before congestion happens.

- **Traffic shaping:** regulate the rate/pattern at which traffic is injected into the network - leaky bucket and token bucket (below).
- **Admission control:** for connection-oriented networks - a new virtual circuit is admitted only if the network can carry it without degrading existing ones (negotiated at setup).

Closed-loop (control): monitor -> detect -> convey -> adjust. Routers watch queue lengths / utilization; when a threshold is crossed they inform sources; sources then slow down (covered below and in TCP).

3. Congestion Control in Datagram Subnets

- **Warning bits:** routers set a special bit in the packet header when congestion is building; the destination echoes the warning back to the source in its reply, and the source slows down.
- **Choke packets:** the congested router sends an explicit control packet directly to the source ('slow down to X rate'). Sources that ignore repeated choke packets get dropped.
- **Hop-by-hop choke packets:** each router on the path applies the relief immediately (relieves congestion faster along the path, at the cost of more control packets).

Load shedding

- When buffers are full, routers must drop packets - choosing **which** to drop is load shedding.
- **Wine vs milk policy:** drop old packets (wine gets better with age) or fresh ones (old milk goes bad first - VoIP prefers dropping old audio).
- **RED (Random Early Detection):** drop packets randomly **before** buffers are completely full, so all sources slow down together gently instead of one flow losing everything.

Jitter control

- **Jitter** = variation in end-to-end packet delay. Even with low average delay, high jitter ruins real-time traffic (VoIP, video).
- Fix: **buffer packets at the receiver** and play them out at constant intervals - smooths delay variation; trade-off is added fixed delay.

4. Leaky Bucket Algorithm

- **Analogy:** a bucket with a small hole at the bottom. Water (packets) can be poured in at any rate, but it leaks out at a constant rate; if the bucket overflows, the excess is lost.
- **Working:** a host sends packets into a queue (the bucket); a scheduler releases exactly one packet every fixed time interval (constant output rate), regardless of how bursty the input is.
- **Effect:** converts a **bursty stream into a perfectly even stream** at the leak rate.
- **Properties:** output rate is always constant; idle periods cannot be 'saved up' - bursts are lost; simple queue with a timer; may waste capacity when the network is idle.

5. Token Bucket Algorithm

- **Working:** tokens are generated at a constant rate (rho) into a bucket of capacity C. To send a packet of size S, the host must remove S tokens; if not enough tokens are available, the packet waits (or is dropped).
- **Effect:** long-term rate is limited to the token rate, but **bursts up to bucket capacity are allowed** - unlike leaky bucket.
- **Idle hosts accumulate tokens** (up to C), so saved capacity can be used in a later burst.
- **Maximum burst length:** S = C / (M - rho), where M is the maximum output rate of the link.

Worked example

Bucket capacity C = 500 KB, tokens generated at rho = 10 MB/s, maximum output rate M = 50 MB/s. Maximum burst length S = C / (M - rho) = 0.5 MB / 40 MB/s = **12.5 ms**.

Leaky Bucket vs Token Bucket

Point	Leaky Bucket	Token Bucket
Bucket holds	Packets (queue)	Tokens
Output rate	Strictly constant	Average = token rate; short bursts up to capacity allowed
Idle sender	Gains nothing; input bursts are lost	Accumulates tokens (up to capacity) for later
Network capacity	Wasted when idle	Used efficiently when idle
Typical use	Hard rate limiting	Traffic shaping allowing controlled bursts

6. Tunnelling

- **Definition:** to move a packet through a network that speaks a different protocol, the whole packet is placed as **data** inside the payload of another packet; at the far end it is extracted (decapsulated) and forwarded normally.
- **Uses:** carrying IPv6 packets across IPv4 networks (6to4); VPNs - private IP packets wrapped in public Internet packets, encrypted; multicast through unicast-only networks; connecting dissimilar WANs.
- Like putting a letter inside a bigger envelope so a foreign postal system can carry it.

7. Internetworking and Internetwork Routing

- **Internetwork (internet):** a collection of networks (LANs, WANs) connected by routers/gateways, appearing as one network to hosts.
- **Problems when routing between different networks:** different addressing schemes, different MTUs (needs fragmentation), different protocols/quality of service, security and administration - the network layer hides all this from the transport layer.
- Routers forward packets hop-by-hop across network boundaries; routing protocols (RIP, OSPF, BGP) exchange reachability so every router knows where to send traffic for distant networks.
- Gateways translate between completely different protocols/architectures (stronger than a router, which connects networks sharing the same protocol family).

8. Fragmentation

- Each network (link) has a **Maximum Transmission Unit (MTU)** - the largest frame payload it can carry. A datagram bigger than the MTU must be split.
- IPv4: routers along the path fragment when needed; **reassembly only at the final destination** (so fragments can take different routes).
- IPv6: routers never fragment - if a packet is too big, the router sends back an ICMPv6 'packet too big' message and the SOURCE must fragment.
- **Fields used:** Identification (all fragments of one datagram carry the same value), MF (More Fragments - 1 on all but the last), DF (Don't Fragment - if set and packet too big, it's dropped + ICMP error), Fragment Offset (position of this fragment in the original data, measured in **units of 8 bytes**).

Worked example

A 4000-byte IP datagram (20-byte header + 3980 bytes data) must cross a network with MTU 1500 bytes. Max data per fragment = 1480 (must be a multiple of 8).

Fragment	Total length	Data bytes	MF	Fragment offset (bytes / 8)
1	1500	1480	1	0 (0 / 8)
2	1500	1480	1	185 (1480 / 8)
3	1040	1020	0	370 (2960 / 8)

- Check: 1480 + 1480 + 1020 = 3980 data bytes. Offsets are 0, 185, 370 (each offset = previous offset + previous data size / 8).
- The destination uses Identification + MF + offset to reassemble in order; the timer discards fragments if some never arrive.

Last-Minute Revision - Numbers and One-Liners

- IPv4 = 32 bits; IPv6 = 128 bits; IPv4 header 20-60 B; IPv6 header fixed 40 B; max IPv4 datagram = 65,535 B.
- Default masks: A /8 (255.0.0.0), B /16 (255.255.0.0), C /24 (255.255.255.0).
- /26 = 62 hosts, /27 = 30 hosts, /30 = 2 hosts; block size = 256 - mask octet.
- RIP = distance vector (metric = hop count, max 15); OSPF = link state (Dijkstra).
- Count-to-infinity fixed by: split horizon, poison reverse, hold-down timers.
- Optimal hierarchy levels = ln n routers.
- DHCP DORA: Discover, Offer, Request, Acknowledge; UDP ports 67 (server) / 68 (client).
- VLAN = IEEE 802.1Q tag = 4 bytes; each VLAN = separate broadcast domain.
- Leaky bucket = constant rate; token bucket = average rate + bursts (S = C / (M - rho)).
- Fragment offset units = 8 bytes; IPv4 reassembly at destination only; IPv6 source fragments only.
- Protocol field: 6 = TCP, 17 = UDP, 1 = ICMP.

Most likely exam questions from this unit

- Explain the IPv4 header field by field (draw the diagram).
- Subnet a given network into N subnets / for N hosts - full table with ranges and broadcast.
- Differentiate IPv4 and IPv6; explain IPv6 addressing notation and header.
- Explain distance vector routing; illustrate count-to-infinity and its remedies.
- Explain link state routing and its five steps; compare with distance vector.
- Run Dijkstra's algorithm on a given graph (labelled steps).
- Explain DHCP with the DORA sequence; explain VLAN and its advantages.
- Compare leaky bucket and token bucket (with the burst-length formula).
- Explain fragmentation with a worked example (MTU calculation).
- What is congestion? Explain congestion control in datagram subnets: choke packets, load shedding, jitter control; explain tunnelling.

All the best for your mid-sems! If you want, I can also make a one-page quick-revision sheet or a practice question set with answers for Unit 2.